

Reproducing a Weekend Transolver for OpenRadioss Bumper-Beam Crash Data with PINNeAPle

100 OpenRadioss simulations, a Latin-hypercube DoE over two gauges, and a slice-attention neural operator — on a laptop

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Abstract

We reproduce, with the PINNeAPle toolchain, a public write-up in which a Transformer-based neural operator (a simplified Transolver) is trained on OpenRadioss simulations of a bumper beam to predict the displacement time history of every node. We re-ran the full pipeline: the OpenRadioss 'Bumper Beam' example, two linked shell gauges as design variables (DP1000 and DP600, $\pm 30\%$ of nominal), a 100-point Latin hypercube with the same seed and bounds, 100 explicit simulations (100 ms, ANIM output converted to VTK) run in Docker on Apple silicon, consolidation into HDF5 with the write-up's schema, and the write-up's architecture and loss (field MSE + $2\times$ the history of node 1806). New code needed for this — an OpenRadioss bridge (deck editing, Docker runner, ANIM/VTK reader) and the slice-attention model — was added to PINNeAPle. On the write-up's ten held-out runs the reproduced model reaches a node-1806 RMSE of 0.968 mm ($R^2 = 0.952$) and a full-field relative L2 error of 0.033. This is consistent with the write-up's claim that the predicted curves track the solver closely.

Keywords: OpenRadioss; crash simulation; surrogate model; Transolver; neural operator; design of experiments; reproducibility; PINNeAPle

Reproduction scope

- Reproduced exactly: model, solver, deck, design variables, bounds, LHS seed (SciPy qmc, seed 42), held-out test IDs, network block and loss weighting.
- Not specified in the write-up and chosen here: optimiser settings, epochs/early stopping, validation split, displacement component (u_x , inferred from the node-1806 curves described), normalisation constants. These are the most likely sources of difference.
- Our OpenRadioss build (latest-20260728, aarch64) and scipy version may differ from the write-up's.

1 Pipeline

- Model: OpenRadioss 'Bumper Beam' example (Altair, CC BY-NC 4.0): DP600 crash boxes and DP1000 beam, shell elements, 5 m/s initial velocity against a rigid pole, 100 ms. The engine deck was changed only to request ANIM displacement output every 1 ms (101 frames).
- DoE: DV1 (DP1000, /PROP/SHELL/2 and /7) $\in [1.26, 2.34]$ mm and DV2 (DP600, /PROP/SHELL/1 and /6) $\in [1.54, 2.86]$ mm, LHS $n = 100$, seed 42, via PINNeAPle `sample_parameters`. Thickness fields edited in place with a byte-preserving deck editor (verified by diff).
- Solves: `pinneapple_simulation.external_solvers.openradioss` runs starter, engine and anim_to_vtk in an Ubuntu container, eight in parallel. Energy-balance error reported by the solver: median 0.30%, worst 0.40%.
- Data: 100 runs, 13,882 nodes, 101 frames, HDF5 with inputs/Exp_i (nodes, elements, gauges) and outputs/Exp_i (displacement field, node-1806 history).
- Split: test = the write-up's IDs 5, 15, 16, 19, 30, 33, 37, 83, 88, 96; validation = 10 further fixed runs; training = the remaining 80.

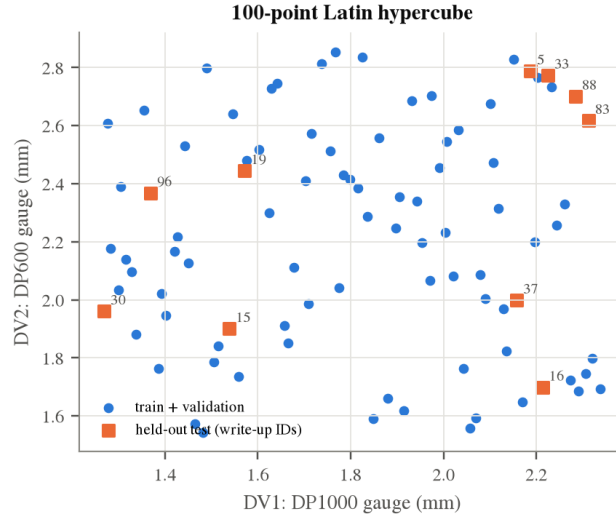


Figure 1. The Latin-hypercube design; squares are the ten held-out test runs of the write-up.

2 Model

TransolverLite (now in PINNeAPPLE as *transolver_lite*): input projection $5 \rightarrow 64$, three slice-attention blocks (8 slices, 4 heads, dropout 0.1), per-node output of 101 values. Node features $[x, y, z \text{ normalised to } [0,1], t_{\text{DP600}}, t_{\text{DP1000}}]$; targets standardised with training-set statistics. Loss: MSE over all nodes and frames + $2 \times \text{MSE}$ on node 1806. Adam, lr 10^{-3} , batch of 2 runs, early stopping on validation. Parameters: 158,461; training 22 min (264 epochs) on Apple-silicon MPS; inference 14 ms per run.

3 Results

Table 1. Held-out test runs (u_x).

Model	node RMSE (mm)	node max lerrl (mm)	node R^2	peak err (mm)	final err (mm)	field rel. L2	shell rel. L2
TransolverLite (reproduction)	0.968	7.163	0.952	0.593	2.844	0.033	0.028

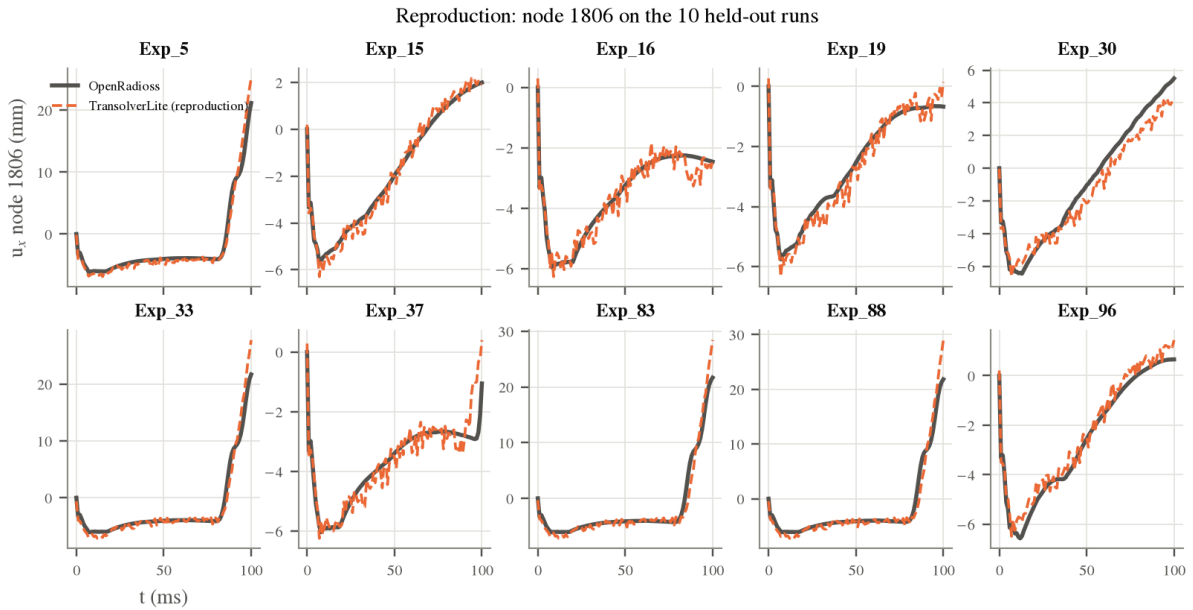


Figure 2. Node-1806 displacement history on the ten held-out runs: OpenRadioss (solid) vs reproduced model (dashed).

Per-run node RMSE ranges from 0.30 to 1.55 mm (worst: Exp_88). The write-up singled out Exp_30 as its only visible deviation; in our reproduction Exp_30 has 0.81 mm.

4 Sources of error in the reproduced model

- Scale mismatch in the target: the field contains parts moving by hundreds of millimetres, while node 1806 moves by a few millimetres. With one global normalisation ($\sigma = 74.4$ mm) the node-1806 signal is a small fraction of the loss; the $2\times$ weighting does not fix this.
- Both gauges are broadcast to every node; the network must infer which nodes belong to which gauge from coordinates alone.
- With only 80 training designs and a two-dimensional design space, a high-capacity point-cloud model is data-starved; a classical surrogate on (DV1, DV2) is a strong baseline (see the companion improvement report).

References

- [1] Chavare, S. Building a ground-up Transolver on OpenRadioss data. Medium (accessed September 2026).
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